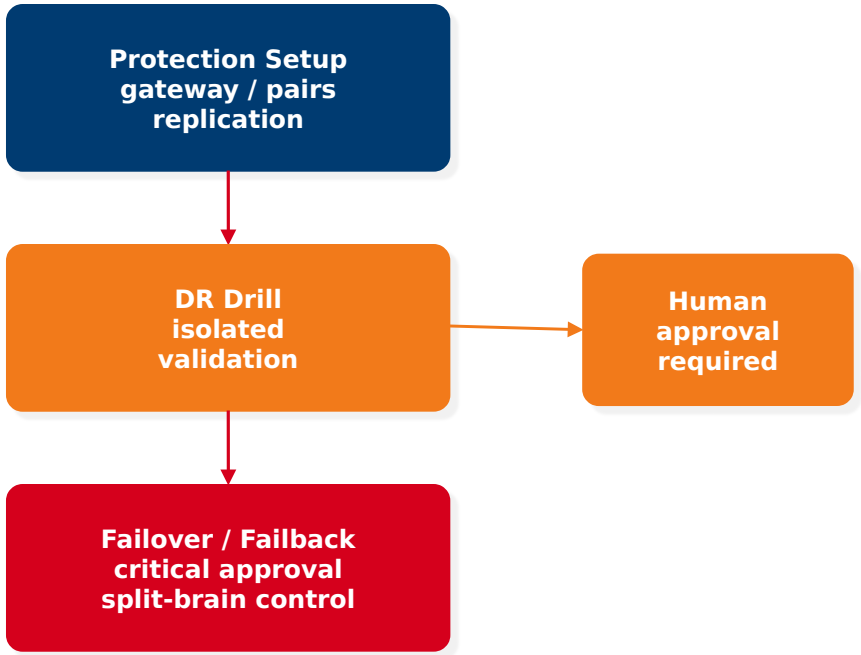
AI-Guided Huawei Cloud Scenario

SDRS Cross-Region DR

AI-guided disaster recovery runbook using supervised console execution and three operational skills

EXPERIMENTAL · CRITICAL RISK · Manual Console



Scenario Flow at a Glance

The scenario README orchestrates the skill, approval gates, evidence and completion criteria.

1 **Protect**

setup and replication readiness

2 **Drill**

isolated recovery validation

3 **Failover**

planned/unplanned critical gate

4 **Failback**

reprotect, sync, return

Primary mechanism
Manual Console

Scenario maturity
EXPERIMENTAL
Risk: CRITICAL RISK

Control model
Discover before create. Verify after every step. No write without explicit approval.

Status
EXPERIMENTAL

Risk
CRITICAL RISK

Core
Manual Console

Control
approval

How to Use AI Safely for SDRS Cross-Region DR

1

What AI Does

Plans, analyzes, prepares runbooks, checks dependencies and explains risks. It does not replace expert review.

2

What to Provide

Use anonymized inventories and approved metadata. Never provide real credentials, keys or customer secrets.

3

How to Authorize

Approve each write, cutover, restore, failover or service-impacting action explicitly.

4

When to Stop

Stop on ambiguity, unsupported region/service, unknown state, missing rollback or unclear owner.

AI accelerates planning and evidence collection; it does not remove the need for security review, rollback planning, and human approval.

Your SDRS Cross-Region DR Scenario

Your Starting Point and Goal

AI must understand the business objective, not just execute isolated commands.

Your Starting Point
Critical workloads require cross-region disaster recovery planning with SDRS concepts, but no local SDRS CLI/MCP exists.

Your Goal
Guide protection setup, DR drill, failover and failback with explicit human approvals and evidence.

Why AI-Assisted
AI structures readiness, approvals, transition contracts, validation and split-brain prevention without pretending automation exists.

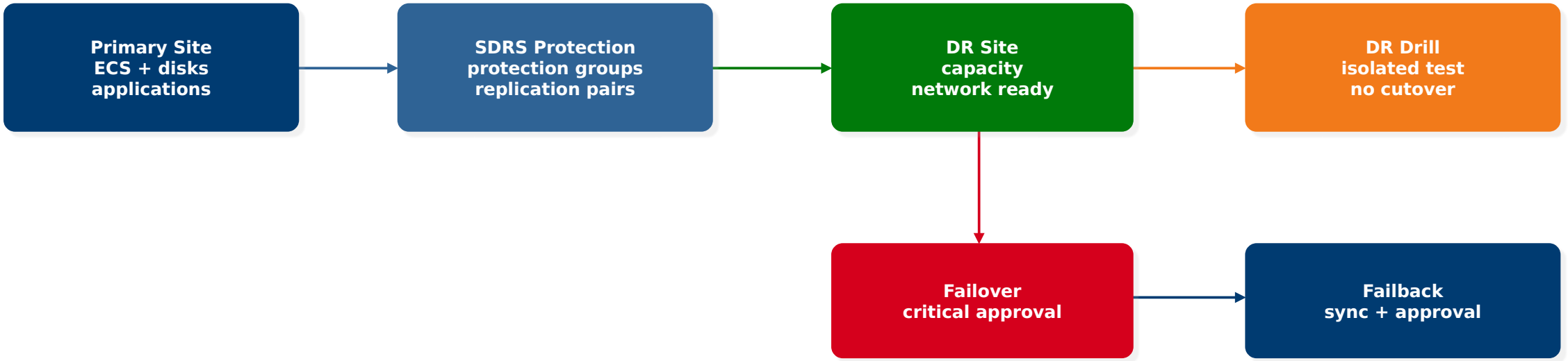
Status
EXPERIMENTAL

Risk
CRITICAL RISK

Mechanism
Manual Console

Finish
evidence

SDRS Cross-Region DR Architecture



Result: proceed only after prerequisites, approvals, and verification evidence are present.

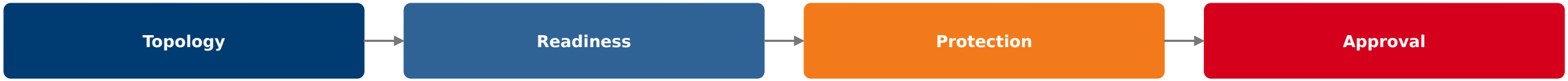
Step 1: Ask AI to Discover and Prove Readiness

Validate SDRS Topology Before Any Action

Example prompt
"Validate SDRS service availability, region pair, source/DR infrastructure, gateway requirement, RPO/RTO targets and approval owner."

AI should know
How to distinguish protection setup, DR drill, failover, reverse reprotection and failback responsibilities.

Verify
No fake hcloud SDRS commands. No automatic DNS. No automatic deletion. Stop on unsupported topology or split-brain uncertainty.



Stop conditions: missing target, ambiguous resource, unsupported state, missing owner, or unclear rollback.

Step 2: Ask AI to Prepare the Controlled Plan

Three Skills, Three Risk Boundaries

Inputs

Plan

Approval

Readiness evidence the agent must produce

- ✓ Protection healthy before drill/failover
- ✓ DR drill optional but recommended
- ✓ Failover requires critical approval
- ✓ Failback has separate plan and approval

Plan elements

Protection Setup

Service/topology validation, gateway, protection groups, pairs and replication readiness.

DR Drill

Isolated test; validates recovery without production DNS/cutover. Drill is not failover.

Failover / Failback

Critical operations: planned/unplanned failover, reprotection, failback and split-brain controls.

Every plan must identify approval owner, verification evidence, and rollback or recovery behavior.

Step 3: Ask AI to Execute One Controlled Phase at a Time

Execution: Controlled Two-Phase Process

Phase 1: Protection + Drill

Configure protection under supervision
Monitor replication state
Run isolated DR drill
Measure RPO/RTO evidence

Ready →

Phase 2: Failover + Return

Planned or unplanned decision
Critical approval
Prevent split brain
Reverse reprotection + failback



On failure: preserve IDs/logs/evidence → classify root cause → resume only from verified checkpoint.

Step 4: Ask AI to Handle Errors and Edge Cases

DR Risk Controls Are Non-Negotiable

R	**Region pair unsupported**	Stop and raise service availability/ticket path.
R	**Replication degraded**	Do not drill/failover without impact analysis and approval.
U	**Uncertain primary state**	Stop to prevent split brain.
D	**DNS request**	Plan only; no automatic DNS changes.
F	**Failback requested**	Requires separate sync, consistency and approval plan.

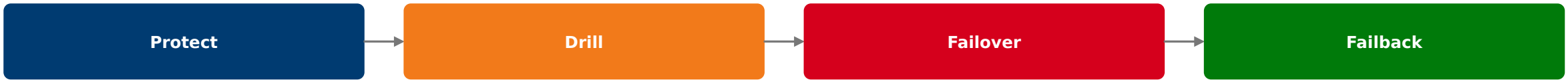
Verify and prevent hallucinations: when AI says “fixed”, require concrete evidence — logs, status, checks, and owner validation.

Step 5: Ask AI to Validate Before Closure

Completion Depends on the DR Objective

Do not accept generic “looks good”. Require proof before closure.

- ✓ Protection setup complete and replication healthy
- ✓ DR drill report proves isolated recovery
- ✓ Failover result includes RPO/RTO and app validation
- ✓ Reverse reprotection state recorded
- ✓ Failback validated with split-brain prevention



Equip Your AI Assistant

What the AI Needs to Execute the Scenario

Start from scenario README; load only the required skill for the current phase.

Scenario README

Scenario README orchestrates transitions and prevents drill/failover/failback confusion.

Operational Skill

Three SDRS SKILL.md files isolate risk boundaries and require explicit approvals.

Shared Components

Capability builder is required for future SDRS automation; candidate tools remain NOT_IMPLEMENTED.

Rule of thumb: do not load every skill; identify the current phase, load the relevant SKILL.md, execute one controlled step, verify, then return to the scenario README.

Pilot to Production Journey

Recommended SDRS DR Journey

This scenario is complete only when its objective-specific validation criteria are met and closure is approved.

